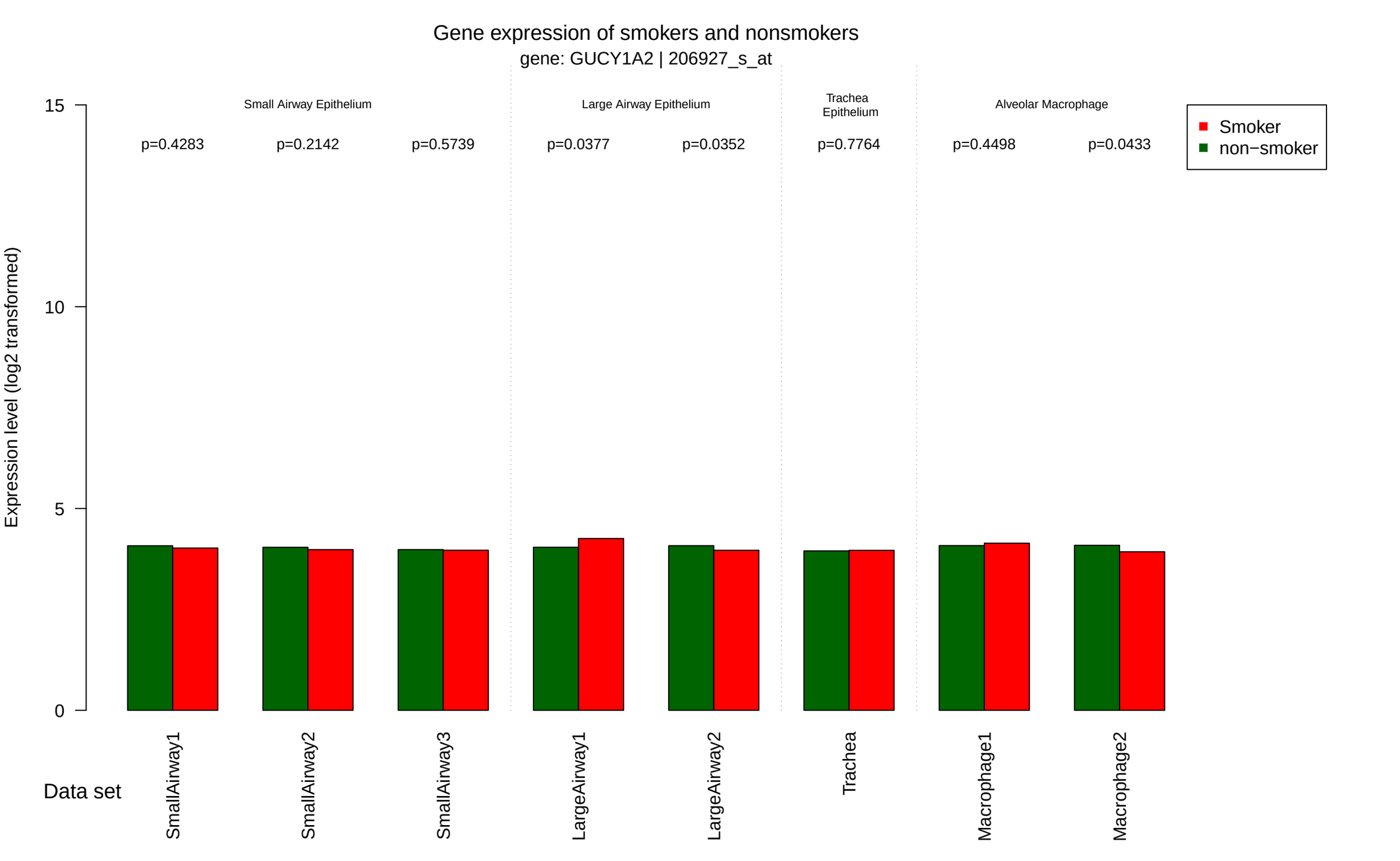
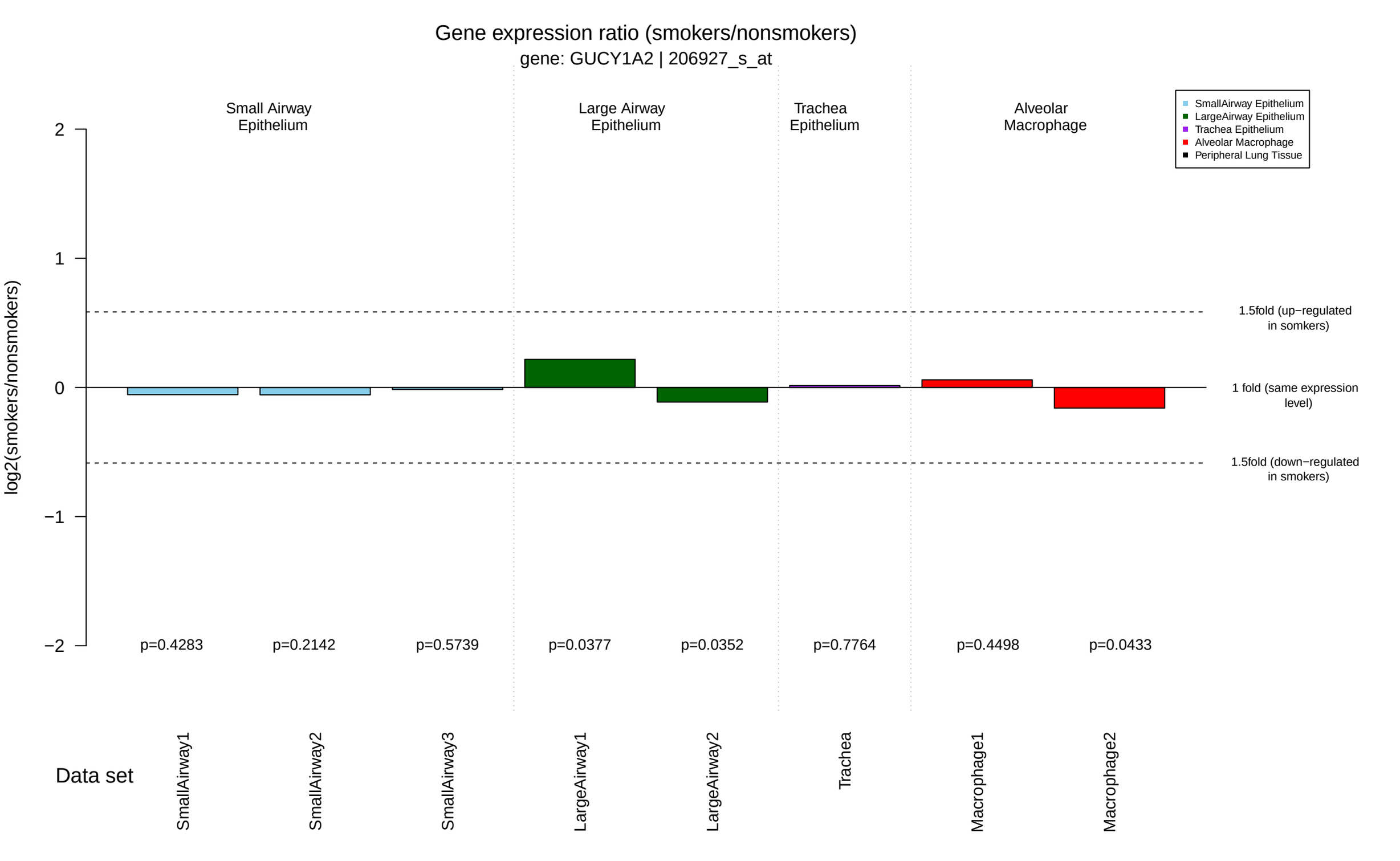
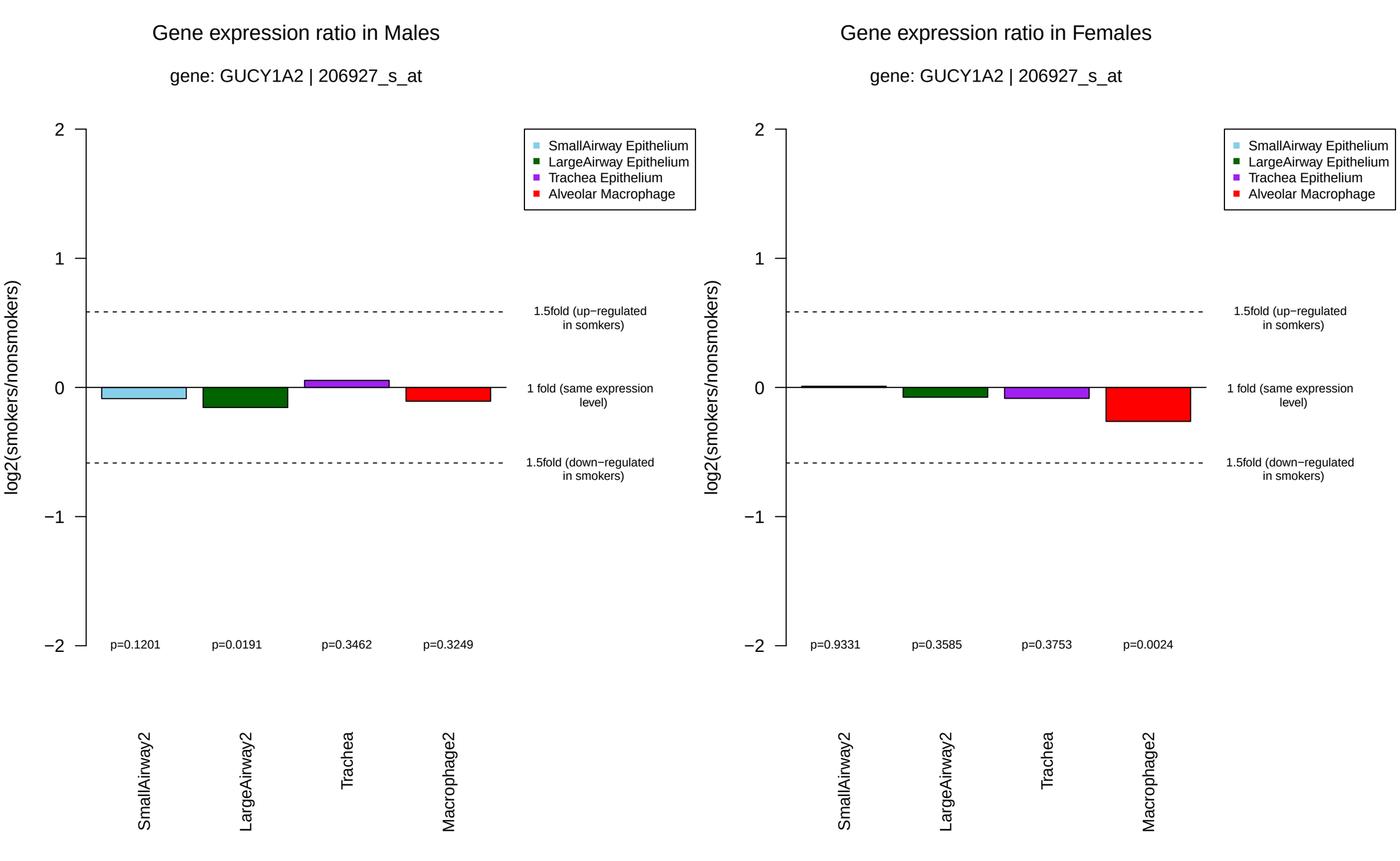
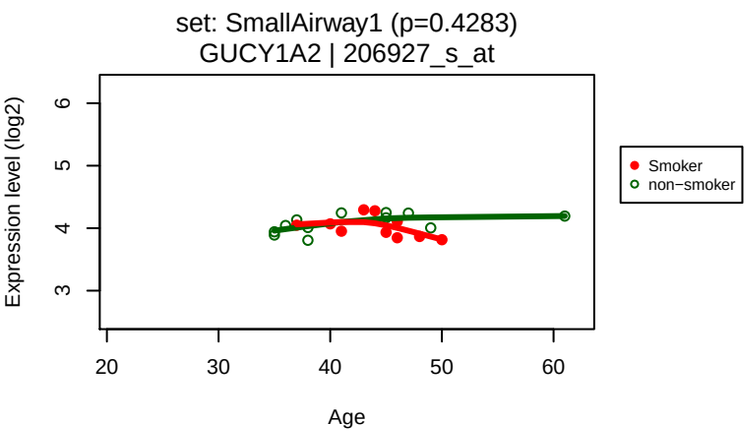
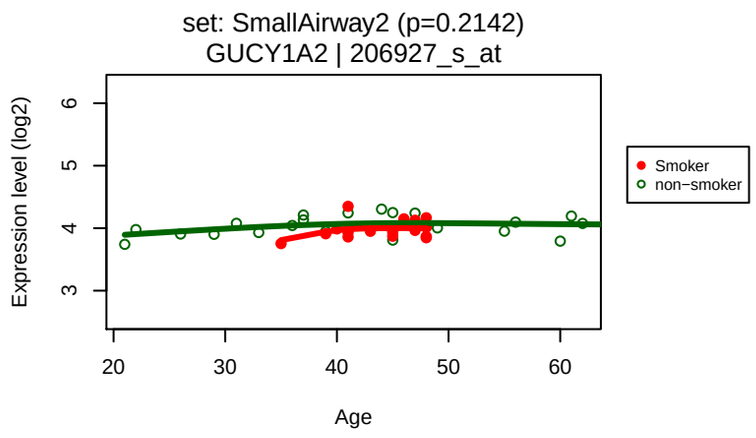
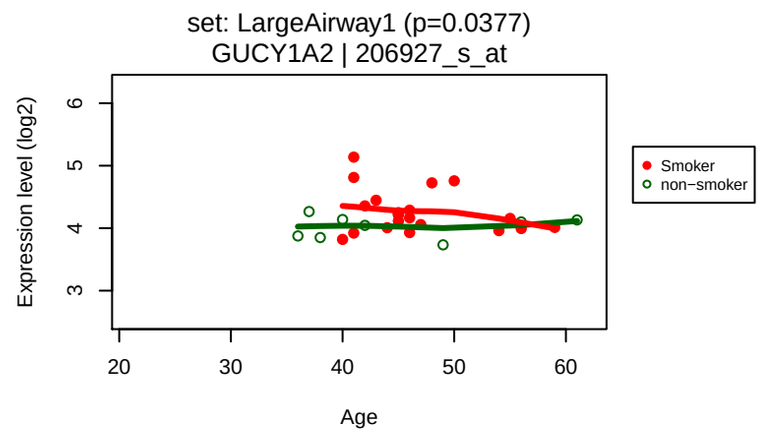

Gene expression of smokers and nonsmokers
gene: GUCY1A2 | 206927_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4283
p=0.2142
p=0.5739
p=0.0377
p=0.0352
p=0.7764
p=0.4498
p=0.0433
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

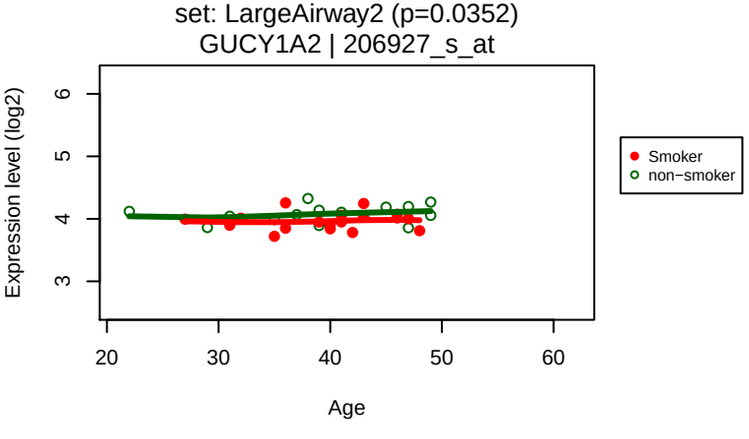

Gene expression ratio (smokers/nonsmokers)
gene: GUCY1A2 | 206927_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4283
p=0.2142
p=0.5739
p=0.0377
p=0.0352
p=0.7764
p=0.4498
p=0.0433
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

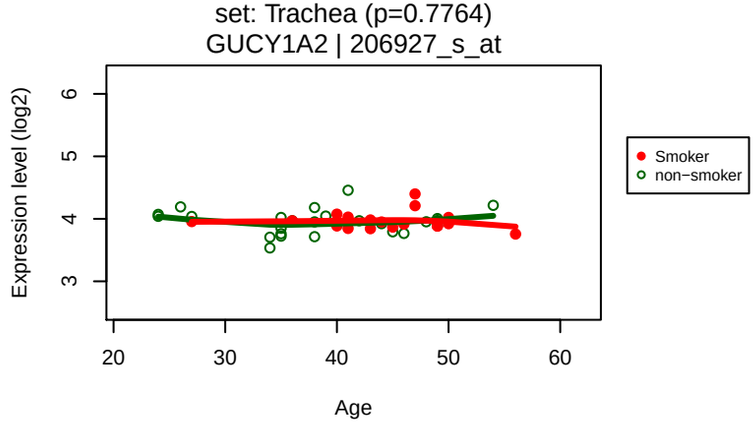

Gene expression ratio in Males
gene: GUCY1A2 | 206927_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1201
p=0.0191
p=0.3462
p=0.3249
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GUCY1A2 | 206927_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9331
p=0.3585
p=0.3753
p=0.0024
SmallAirway2
LargeAirway2
Trachea
Macrophage2

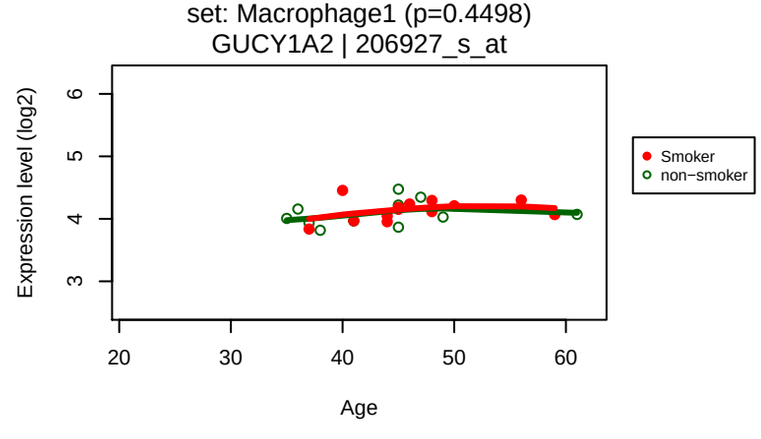

set: SmallAirway1 (p=0.4283)
GUCY1A2 | 206927_s_at
Expression level (log2)
Age
Smoker
non−smoker

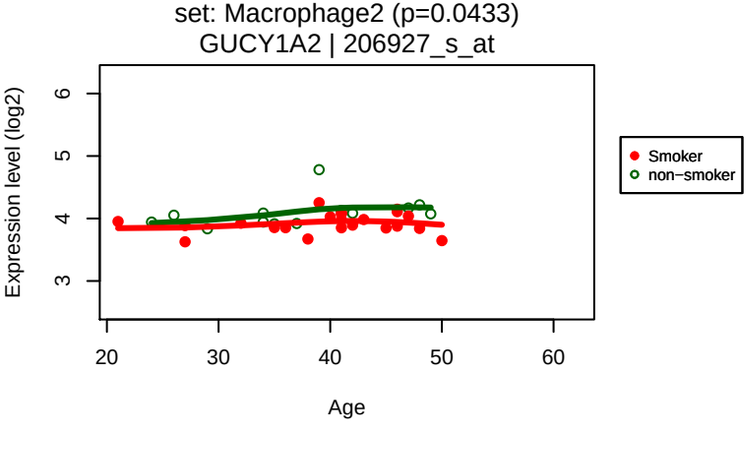

set: SmallAirway2 (p=0.2142)
GUCY1A2 | 206927_s_at
Expression level (log2)
Age
Smoker
non−smoker

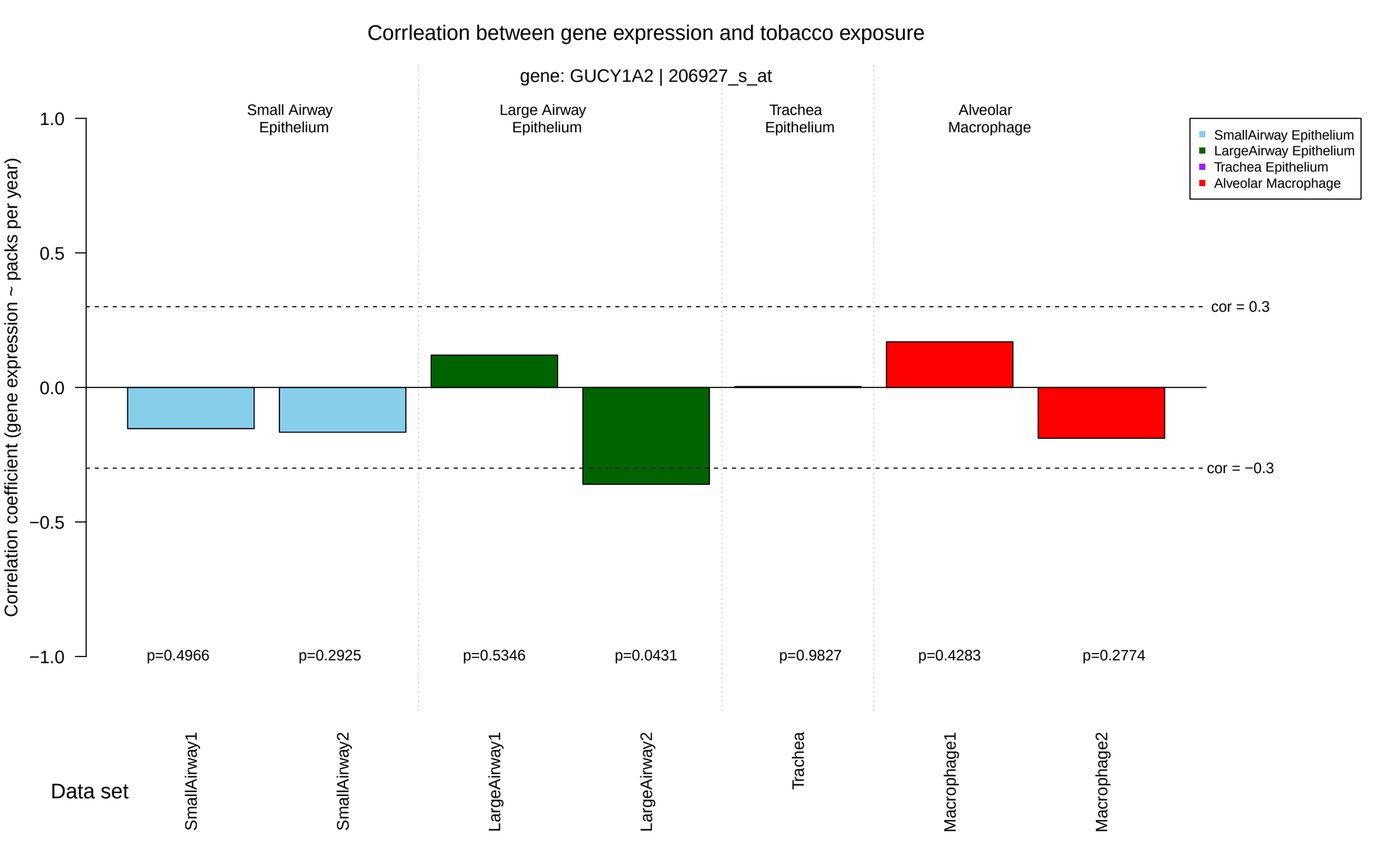

set: LargeAirway1 (p=0.0377)
GUCY1A2 | 206927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0352)
GUCY1A2 | 206927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7764)
GUCY1A2 | 206927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4498)
GUCY1A2 | 206927_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0433)
GUCY1A2 | 206927_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GUCY1A2 | 206927_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4966
p=0.2925
p=0.5346
p=0.0431
p=0.9827
p=0.4283
p=0.2774
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2